

VoiceCogs: Interlocking Concurrent Voices for Separable Compressed Browsing with Screen Readers

Jeongwon Choi
choijw@postech.ac.kr
POSTECH

Pohang, Gyeongbuk, South Korea

Inseok Hwang
i.hwang@postech.ac.kr
POSTECH

Pohang, Gyeongbuk, South Korea

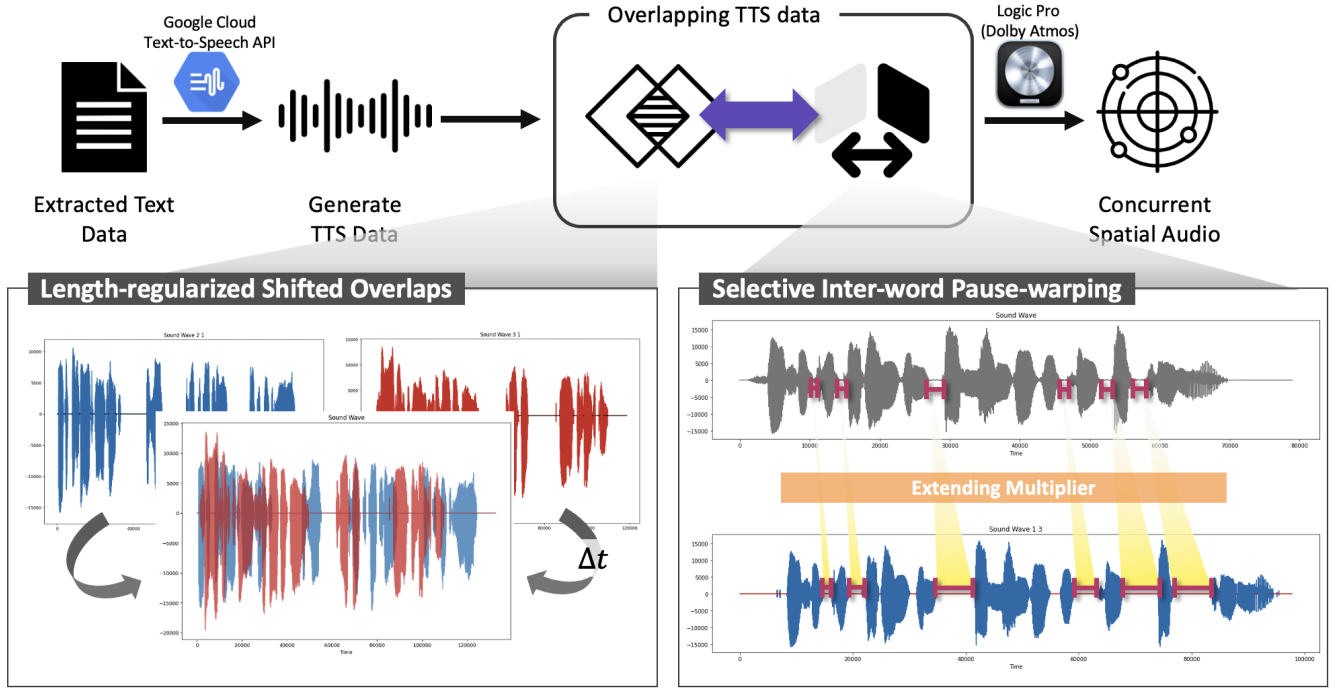


Figure 1: VoiceCogs System Flow Chart

ABSTRACT

Ensuring universal accessibility to information cannot be overstated. Unlike visual readers, however, screen reader users are given inefficient and restricted channels to acquire the given information. In particular, we focus on the initial step of information acquisition – *quickly* scanning the overall structure of a textual document so that the reader makes an informed decision about where to jump and read the details. While this step is inherently quick for visual users, screen reader users passively listen to the slow, sequential list of items read aloud. To close this gap, we call for a technique that accelerates screen reader users’ scanning process. Our system, VoiceCogs, takes multi-itemed text sources and synthesizes audio

that concurrently plays multiple text-to-speech from a respective text source while facilitating the discernibility of individual sources. To this end, we devise and implement two interlocking techniques to minimize phonetic interferences between concurrent speeches.

CCS CONCEPTS

• **Human-centered computing** → **Accessibility systems and tools**; *Sound-based input / output*; *Interactive systems and tools*.

KEYWORDS

concurrent speech; spatial audio; screen reader; accessibility

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1 INTRODUCTION

Upon seeing a newspaper or webpage, in which order would you browse the contents? Perhaps you would capture the page structure by quickly scanning the titles and headlines, then jump on the text block that interests you. Here, your vision gets the initial scanning done very fast; the concurrent presence of spatially distributed multiple visual items hardly interferes with each other [22]. The proliferation of VR is making the viewing space even larger [7]. You can even control the scanning order by leveraging the spatial proximity of text blocks.

Unfortunately, people with visual impairment who rely on screen readers neither have such fast scanning nor controllability of the scanning order. In general, listening is much slower than viewing for grasping the content [8, 9, 16, 17, 19]; it takes much more time for screen reader users to go through the list of titles and headlines than visual readers to do the same by eyes. Furthermore, screen reader users have little freedom to deviate from the order that the screen reader reads aloud along the page structure. Eventually, screen reader users have inherent disadvantages in terms of time-efficiently browsing structured texts.

We propose VoiceCogs, a screen reader system that facilitates *compressed browsing* of structured texts by overlapping multiple text phrases while keeping the aural discernibility high. As it is well-known that simultaneously speaking while another is speaking is discouraged for social and intelligibility reasons [14, 15, 18], we develop VoiceCogs to address the intelligibility challenges originating from overlapping.

VoiceCogs interlocks multiple text phrases in a way that *minimizes their syllabic overlaps*. In particular, we devise two techniques: (1) length-regularized shifted overlaps, and (2) selective inter-word pause-warping. VoiceCogs firstly adjust the time offset of each text phrase to be read aloud so that the text phrases are overlapped in time but little coincide in the syllabic levels, while regularizing the time offset to control the total length of the overlapped speech. To reduce syllabic collision even fewer under a given total length limit, we selectively warp inter-word gaps where otherwise the syllables from different phrases momentarily overlap. To further improve the discernibility of overlapped voices, VoiceCogs additionally employs spatial audio [1, 6, 11, 12, 20] and varying individual voices.

In this paper, we present the key algorithms and the implementation of VoiceCogs, followed by the evaluation results from a pilot study with 15 participants.

2 RELATED WORK

Screen readers are widely used to provide an alternative user interface to textual contents for people with visual impairment [10, 21].

The human cognitive system has the ability of selective attention, i.e., focusing on the specific content while simultaneously accommodating multiple external sounds at the same time, also known as Cocktail Party Effect [4, 5] where people can selectively listen to and understand one talker in a multi-talker situation. Additionally, it is known that the ability to separate concurrent speeches can be improved by learning [23]. The design rationale of VoiceCogs is analogous to such inherent human ability of selective attention

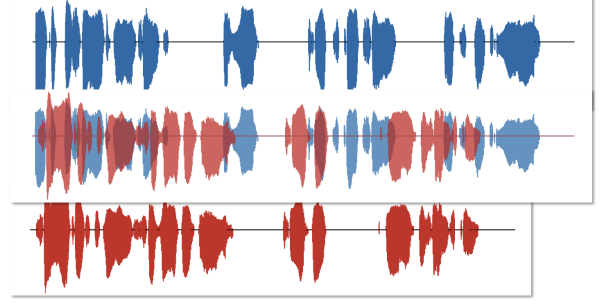


Figure 2: Length-regularized Shifting Example

in multi-talker situations, on top of which we add microscopic temporal alterations to each independent speech source.

3 VoiceCogs

3.1 Design

Figure 1 depicts the overall flow of VoiceCogs system. First, we extract the target page’s titles and headings, followed by TTS generating [2] the audio that reads aloud individual text phrases. Then, we apply multiple iterations of *length-regularized shifted overlaps* and *selective inter-word pause-warping* to reach an optimal balance between the syllabic overlaps and the total length of the overlapped speech. Finally, we use a DAW (Digital Audio Workstation) to synthesize the spatial audio effect that mimics the independent voices spatially separated as desired.

3.2 Optimizing Voice Overlaps

In the process of overlapping multiple voices, our algorithm finds an appropriate time offset Δt for each voice source to minimize syllabic overlaps with other concurrent voices, so that the discernibility of each voice can be improved. Formally, our algorithm solves the following optimization problem, Eq. (1):

$$\begin{aligned} \underset{\Delta t}{\operatorname{argmin}} \quad & \left[\frac{1}{N} \sum_{t=0}^N \{E_1(t) + E_2(t + \Delta t)\} + \lambda |\Delta t| \right] \\ \text{where } \quad & \begin{cases} N & : \text{total length of audio} \\ s_i(t) & : \text{audio signal of } i\text{-th voice} \\ E_i(n) = \sum_{t=0}^n s(t)^2 & : \text{energy of } i\text{-th voice} \end{cases} \end{aligned} \quad (1)$$

To further increase the effectiveness of the algorithm mentioned in Eq. (1), we run multiple iterations of alternating ‘length-regularized shifted overlaps’ and ‘selective inter-word pause-warping’. As shown in the Figure 2, ‘length-regularized shifted overlaps’ is a shifting method that minimized the overlap between the concurrent audios. And ‘selective inter-word pause warping’ functions to warp the pause between words without losing context and identification. We empirically set the lower- and upper-bounds of inter-word pauses. Then, at each iteration, we apply small perturbations to the inter-word pauses within the bound, followed by re-running the optimization problem (1).

Condition	# of subpage titles	# of unique voices	# of concurrent audio	Spatiality	Length-regularized shift	Inter-word pause warping	mean Likert Scale (μ)	std (σ)
C1.1	2	2	2	X	X	X	3.87	1.06
C1.2	2	2	2	O	X	X	4.47	0.74
C1.3	2	2	2	O	O	X	3.87	0.83
C1.4	2	2	2	O	X	O	3.93	1.10
C1.5	2	2	2	O	O	O	4.33	0.72
C2.1	4	4	4	X	X	X	1.00	0.00
C2.2	4	4	4	O	X	X	1.73	0.70
C2.3	4	4	4	O	O	X	3.00	0.85
C2.4	4	4	4	O	X	O	2.47	0.83
C2.5	4	4	4	O	O	O	3.20	0.86

Table 1: Conditions for tasks browsing subpage titles.

Condition	# of contents	# of unique voices	# of concurrent audio	Spatiality	Length-regularized shift	Inter-word pause warping	mean play time (μ)	std (σ)
C3.1	8	2	0 (linear)	X	X	X	16.00	7.41
C3.2	8	2	2	O	X	X	9.07	4.13
C3.3	8	2	2	O	O	O	10.8	5.17

Table 2: Conditions for tasks browsing an auto-attendant phone system's menus.

3.3 Implementation

We use Google Cloud Text-to-Speech API as the TTS. With Speech Synthesis Markup Language (SSML), we control the detailed elements of generated voice, such as pronunciation, volume, pitch, emphasis, and rate. We employ Apple Logic Pro [3] as our DAW (Digital Audio Workstation), where we apply Dolby Atmos plug-in – one of the most popular surround sound technologies. It provides a virtual space where each independent sound source can be placed at a specific location.

4 PILOT STUDY

To evaluate the discernibility of the VoiceCogs-generated concurrent voices, we designed a pilot study with a group of voluntary participants and itemized lists of text highlights (e.g., section headings, subpage titles) from a structured text. We recruited a total of 15 university students (F=3, M=12) with an age range of 20 to 29 (mean=23.5, std=2.7). Based on the literature that reported no significant difference in discriminating concurrent audio between the visually impaired and the non-impaired [12], we conducted the pilot study with the participants who are visually non-impaired.

We designed two sets of sample tasks – (1) browsing the subpage titles listed in a university bulletin page, and (2) browsing the menus of an auto-attendant phone system. In each task, we varied the number of unique voice tones and concurrent voices, as well as the ablations of individual techniques that we proposed. Table 1 and 2 organize the detailed experimental conditions and results in each task.

We received responses about the intelligibility of the concurrent voices on a 5-point Likert scale. The mean value in Table 1 implies the degree of discrimination of overlapping voices, and Table 2 implies the average time spent browsing. The detailed results for each task are shown in the Figure 3. The pilot study has shown that hearing multiple menus concurrently with VoiceCogs reduced

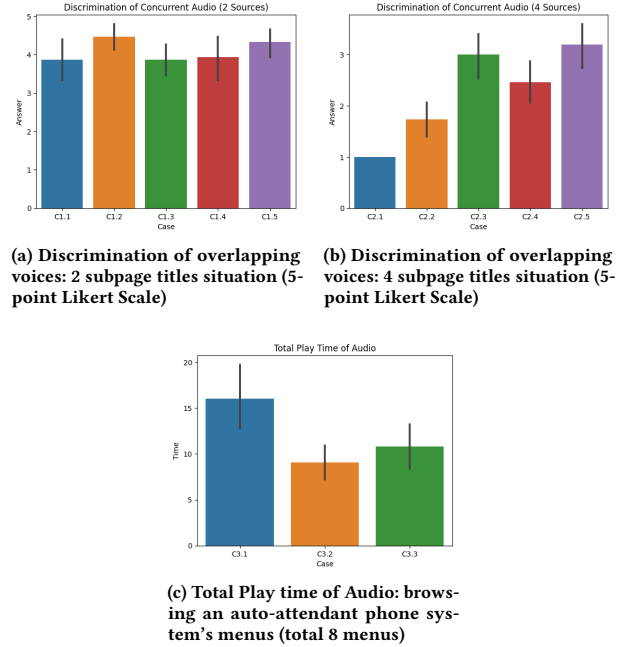


Figure 3: Pilot Study Result

the average time required to find the desired menu. Also, applying overlapping techniques increases the discrimination, especially in four audio source situation.

P11 said, “The sounds don’t feel completely overlapping. It felt more listenable than the previous ones.” after hearing audio that applied full techniques: pitch of voice, spatiality, length-regularized shift,

and inter-word pause-warping. Also, P9 mentioned improving their ability to identify simultaneous speech through practicing.

5 CONCLUSION AND FUTURE WORK

In this research, we developed VoiceCogs – a screen reader system that facilitates *compressed browsing* of structured texts by inter-locking multiple voices of text phrases in a way that minimizes their syllabic overlaps and thereby keeping the aural discernibility high. In this light, we devised two techniques in VoiceCogs, namely length-regularized shifted overlaps and selective inter-word pause-warping. Throughout a pilot study, we obtained the preliminary evaluation results that shed light on the usability and time efficiency of VoiceCogs.

For future works, we can enhance VoiceCogs by taking account of additional dimensions in controlling the syllabic overlaps and evaluating their discernibility, such as phonetic or prosodic metrics. We may also introduce a real-time adaptation of independent voice sources' virtual spatial placement to reflect the user's potential real-time interest changes. When there are too many items to be compressed into one overlapping group, we may adopt a probabilistic model to find a tradeoff between compression density and intelligibility [13]. Also, we need to diversify the demographics of our participant pool, e.g., to include the visually impaired population.

6 DEMO PLAN

We expect that a screen reader generating concurrent speeches might not be familiar to most users. For easy understanding, listening directly to our VoiceCogs system is essential. In order to guarantee a clear experience, we will prepare headphones with noise canceling functions.

For a demo, we will provide an interface for participants to compare the degree of discrimination of the synthesized overlapping voices. First, participants will be asked to select sample sentences to be overlapped. Based on these samples, our system synthesizes speeches applying each technique one by one. Participants will then listen to the synthesized speech using headphones and evaluate the improvement in scanning efficiency and the intelligibility of the speech between outputs.

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